

GMEX: Gulf of Mexico US survey data processing summary

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General info

This document presents the cleaning code and summary of the Gulf of Mexico (US) bottom trawl survey provided by Jeff Rester, Coordinator - Gulf States Marine Fisheries Commission - Habitat Focus Team - Gulf of Mexico Program & David Hanisko, Research Fisheries Biologist, National Marine Fisheries Service, Southeast Fisheries Science Center. It contains annual data from 1982 and up to 2024.

Data cleaning in R

```
#####
#### R code to clean trawl survey Gulf of Mexico
#### Public data Ocean Adapt
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#### Coding: Michelle Stuart, Dan Forrest, Zoë Kitchel November 2021
#####
####Update
####Zoe Kitchel
#### May 4, 2024
####Following issue 47, need to update sum technique to remove duplicates
####Malin Pinsky
#### 10 December, 2025
####Following issue #46, we are updating the GMEX data joins to follow DISMAP approach
#### the previous FishGlob / OceanAdapt code joined these two tables on the incorrect column, which int:
### this is corrected below and now aligns with the DismAP GMEX pipeline (currently on a dev branch: ht
### to make this change we had to update the source file for gmex_bio; the previous link this script po
#### we now have five data objects that need to be merged:
#### gmex_station, *tow, *bio, *spp, *cruise
#### first, we merge tow (haul data) with bio (catch data)
#### these get joined on a shared identifier column, invrecid
#### but previously some rows of bio were dropped because this column was N |> ULL
#### below those are filled in correctly (code from D. Hanisko via C. Gonzales / M. Karp at NOAA) and t:
#####
#Relevant Organizations
#Gulf States Marine Fisheries Commission: https://www.gsmfc.org/seamap-gomrs.php
#Southeast Area Monitoring and Assessment Program Reports:
#https://www.fisheries.noaa.gov/southeast/funding-and-financial-services/
#southeast-area-monitoring-and-assessment-program-seamap

#Helpful reference document
#https://sedarweb.org/docs/wpapers/SEDAR7\_DW1.pdf
#Many different survey events included in the files we pull in
#Most consistent through time are Summer SEAMAP 1987-on and Fall SEAMAP 1988-on
#All other surveys are excluded in the following code

#-----#
#### LOAD LIBRARIES AND FUNCTIONS ####
#-----#
library(tidyverse)
library(lubridate)
library(googledrive)
library(taxize) # for getting correct species names
library(magrittr) # for names wrangling
library(readr)
library(dplyr)
library(PBSmapping)
library(here)
```

```

source("functions/clean_taxa.R")
source("functions/write_clean_data.R")
source("functions/apply_trimming_method1.R")
source("functions/apply_trimming_method2.R")
source("functions/flag_spp.R")
fishglob_data_columns <- read_excel("standard_formats/fishglob_data_columns.xlsx")

# function to calculate convex hull area in km2
#developed from http://www.nceas.ucsb.edu/files/scicomp/GISSeminar/UseCases/CalculateConvexHull/Calcula
calcarea <- function(lon,lat){
  hullpts = chull(x=lon, y=lat) # find indices of vertices
  hullpts = c(hullpts,hullpts[1]) # close the loop
  lonlat <- data.frame(cbind(lon, lat))
  ps = appendPolys(NULL,mat=as.matrix(lonlat[hullpts,]),1,1,FALSE) # create a Polyset object
  attr(ps,"projection") = "LL" # set projection to lat/lon
  psUTM = convUL(ps, km=TRUE) # convert to UTM in km
  polygonArea = calcArea(psUTM,rollup=1)
  return(polygonArea$area)
}

#acts like sum(na.rm=T) but returns NA if all are NA
sumna <- function(x){
  if(!all(is.na(x))) return(sum(x, na.rm=T))
  if(all(is.na(x))) return(NA)
}

#-----#
#### READ IN RAW DATA FILES USING DISMAP METHODS
#### see https://github.com/nmfs-ost/DisMAP/blob/main/data_processing_rcode/code/Compile_Dismap_Current
#### Visit the [Gulf of Mexico]("https://seamap.gsmfc.org/") website
#### Click "Download" the SEAMAP Trawl/Plankton, Bottom Longline
#### Fill in the form ("Scientific Research", "Educational Institution", "Trawl/Plankton Data (CSV)"
#### Unzip the CSV in your downloads folder
#### then copy them into the data folder with the script below (this folder is not tracked by git)
#-----#

# station table
gmex_station <- read_csv(here::here("data", "starec.csv"), col_types = cols(.default = col_character()))
dplyr::select('STATIONID', 'CRUISEID', 'CRUISE_NO', 'P_STA_NO', 'TIME_ZN', 'TIME_MIL', 'S_LATD', 'S_L

gmex_station <- type_convert(gmex_station, col_types = cols(
  # field definitions from SEAMAP Data Structures Version 2, Aug 28, 2014, Gulf States Marine Fisheries
  # https://seamapdata.gsmfc.org/rawl/04%20-%20SEAMAP%20Trawl%20Data%20Structures.pdf
  STATIONID = col_double(), # a unique integer assigned for each entry in STAREC table.
  CRUISEID = col_double(), # a unique integer assigned for each entry in CRUISES table.
  CRUISE_NO = col_double(), # a four character string usually in the format YYYY. Such as 1304 for year
  P_STA_NO = col_character(), # the Pascagoula Station Number. A five character string, using in the fo
  TIME_ZN = col_double(), # a one character field which is a code which represents the time zone of the
  TIME_MIL = col_character(), # a four character field which must be numeric. This field represents sta
  S_LATD = col_double(), # a two character field which is a numeric positive integer and represents lat
  S_LATM = col_double(), # a five character field which is numeric and represents latitude minutes. Fie
  S_LOND = col_double(), # a three character field which is numeric positive integer and represents sta

```

```

S_LONM = col_double(), # a five character field which is numeric and represents starting longitude minutes
E_LATD = col_double(), # a two character field which is a numeric positive integer and represents end
E_LATM = col_double(), # a five character field which is numeric and represents latitude minutes. Five
E_LOND = col_double(), # a three character field which is numeric positive integer and represents end
E_LONM = col_double(), # a five character field which is numeric and represents ending longitude minutes
DEPTH_SSTA = col_double(), # a six character field which must be numeric. This field represents the sta
STAT_ZONE = col_double(), # a five character field which represents the shrimp statistical zone.
MO_DAY_YR = col_date(format = "%Y-%m-%d"), # a date field, which in MM-DD-YYYY format, MM represents
VESSEL_SPD = col_double(), # in the field format XX.X and represents the speed of the vessel in knots
COMSTAT = col_character() # a text comment field, up to 250 characters.
))
names(gmex_station)<-tolower(names(gmex_station))

# tow table. Total counts and weights of finfish, crustaceans and other organisms at specific sample locations
gmex_tow <- readr::read_delim(here::here("data", "invrec.csv"),
                             delim = ',', escape_backslash = T, escape_double = F)
gmex_tow<-type_convert(gmex_tow, col_types = cols(
  # field definitions from SEAMAP Data Structures Version 2, Aug 28, 2014, Gulf States Marine Fisheries
  # https://seamapdata.gsmfc.org/trawl/04%20-%20SEAMAP%20Trawl%20Data%20Structures.pdf
  INVRECID = col_integer(), # a unique integer assigned for each entry in the INVREC table.
  STATIONID = col_integer(), # a unique integer assigned for each entry in the STAREC table.
  CRUISEID = col_integer(), # a unique integer assigned for each entry in the CRUISES table.
  VESSEL = col_integer(), # a unique integer representing the Vessel name, from the VESSELS table.
  CRUISE_NO = col_integer(), # a four character string usually in the format YYXX. Such as 1304 for year
  P_STA_NO = col_character(), # the Pascagoula Station Number. A five character string, using in the f
  GEAR_SIZE = col_integer(), # a three character field which is numeric. This field represents the net o
  GEAR_TYPE = col_character(), # a two character field which represents a gear code. It may be blank or
  MESH_SIZE = col_double(), # a five character field which must be numeric. The field format is XX.XX a
  OP = col_character(), # a one character field which is a code. This code may be blank.
  MIN_FISH = col_integer(), # a four character field which is numeric and integer. The field format is
  WBCOLOR = col_character(), # a one character field which may be blank. This field represents the gros
  BOT_TYPE = col_character(), # a two character field which may be blank. Valid values are: 'B', 'CL', 'C
  BOT_REG = col_character(), # a two character field which may be blank. Valid values are: 'S', 'L', 'O',
  TOT_LIVE = col_double(), # a seven character field which must be numeric. This field contains a numb
  FIN_CATCH = col_double(), # a seven character field which must be numeric. This field contains a numb
  CRUS_CATCH = col_double(), # a seven character field which must be numeric. This field contains a numb
  OTHR_CATCH = col_double(), # a seven character field which must be numeric. This field contains a numb
  T_SAMPLEWT = col_double(), # an eight character field which must be numeric. This field contains a nu
  T_SELECTWT = col_double(), # an eight character field which must be numeric. This field contains a nu
  FIN_SMP_WT = col_double(), # an eight character field which must be numeric. This field contains a nu
  FIN_SEL_WT = col_double(), # an eight character field which must be numeric. This field contains a nu
  CRU_SMP_WT = col_double(), # an eight character field which must be numeric. This field contains a nu
  CRU_SEL_WT = col_double(), # an eight character field which must be numeric. This field contains a nu
  OTH_SMP_WT = col_double(), # an eight character field which must be numeric. This field contains a nu
  OTH_SEL_WT = col_double(), # an eight character field which must be numeric. This field contains a nu
  COMBIO = col_character(), # a two hundred character field used for comments, which may be blank.
  X28 = col_character()
))
gmex_tow <- gmex_tow %>%
  dplyr::select('CRUISEID', 'STATIONID', 'VESSEL', 'CRUISE_NO', 'P_STA_NO', 'INVRECID', 'GEAR_SIZE', 'GEAR_TYPE', 'FIN_CATCH', 'CRUS_CATCH', 'OTHR_CATCH', 'T_SAMPLEWT', 'T_SELECTWT', 'FIN_SMP_WT', 'FIN_SEL_WT', 'CRU_SMP_WT', 'CRU_SEL_WT', 'OTH_SMP_WT', 'OTH_SEL_WT', 'COMBIO', 'X28')
  filter(GEAR_TYPE=='ST') #ST = shrimp trawl (this is what OceanAdapt does too, preserves 90% of tows)

```



```

# biology table. Count totals and weight totals of specific biological catch at a sample location.
gmex_bio <- readr::read_delim(here::here("data", "bgsrec.csv"),
                             delim = ',', escape_backslash = T, escape_double = F)

gmex_bio <- type_convert(gmex_bio, cols(
  # field definitions from SEAMAP Data Structures Version 2, Aug 28, 2014, Gulf States Marine Fisheries
  # https://seamapdata.gsmfc.org/trawl/04%20-%20SEAMAP%20Trawl%20Data%20Structures.pdf
  CRUISEID = col_integer(), # a unique integer assigned for each entry in the CRUISES table.
  STATIONID = col_integer(), # a unique integer assigned for each entry in the STAREC table.
  VESSEL = col_integer(), # a unique integer representing the Vessel name, from the VESSELS table.
  CRUISE_NO = col_integer(), # a four character string usually in the format YYXX. Such as 1304 for year
  P_STA_NO = col_character(), # the Pascagoula Station Number. A five character string, using in the fo
  GENUS_BGS = col_character(), # a seven character field which contains the genus part of the genus/spe
  SPEC_BGS = col_character(), # a six character field which contains the species part of the genus/spec
  BGSCODE = col_character(), # a one character field which contains a bgs code. Valid values are T,E,C,
  BIO_BGS = col_integer(), # a 9 digit field containing a number (biocode) which is based on the genus/
  CNTEXP = col_double(), # CNTEXP is an eight digit numeric field which represents one of two possible
  SELECT_BGS = col_double() # a seven character field which must be numeric. This field contains a numb
))

# cruise table
gmex_cruise <- read_csv(here::here("data", "cruises.csv"), col_types = cols(.default = col_character()))
dplyr::select(CRUISEID, VESSEL, TITLE, SOURCE)
gmex_cruise <- type_convert(gmex_cruise, col_types = cols(
  # field definitions from SEAMAP Data Structures Version 2, Aug 28, 2014, Gulf States Marine Fisheries
  # https://seamapdata.gsmfc.org/trawl/04%20-%20SEAMAP%20Trawl%20Data%20Structures.pdf
  CRUISEID = col_integer(), # a unique integer assigned for each entry.
  VESSEL = col_integer(),
  TITLE = col_character(), # a descriptive title for the survey
  SOURCE = col_character())) # a two digit code for the source (SEAMAP Partner) that collected the data
names(gmex_cruise) <- tolower(names(gmex_cruise))

# species table
gmex_spp <- read_csv(here::here("data", "biocodes_01182023_u1.csv"))
problems(gmex_spp) # seems ok on visual inspection
names(gmex_spp) <- tolower(names(gmex_spp))
gmex_spp <- dplyr::select(gmex_spp,
  # field definitions from SEAMAP Data Structures Version 2, Aug 28, 2014, Gulf S
  # https://seamapdata.gsmfc.org/trawl/04%20-%20SEAMAP%20Trawl%20Data%20Structure
  # note that this table is not actually defined in this publication
  biocode, # the NMFS assigned unique value for each entry. Referred to as "BIOCD
  ciu_biocode, # from Melissa Karp, NMFS: Since multiple changes may have occurre
  taxon, # 13 Dec 2025 Malin Pinsky note: this appears to be the full scientific
  taxonomic) # the taxonomic or scientific name for each entry. 13 Dec 2025 Mali

#-----#
#### REFORMAT AND MERGE DATA FILES ####
#-----#

##Resolve issues

```

```

#Issue 1: Proper way to Merge the Tow (invrec) and bio (bgsrec) tables
# The proper way to link the invrec table to the bgsrec is supposed to use the invrecid
# variable as the primary key. However, the bgsrec table has null invrecid for data collected
# under previous data collection systems. The invrec and bgsrec tables can be linked using
# the vessel, cruise_no and p_sta_no variables as a primary key. Unfortunately, there are
# a series stations where the Oregon II (Vessel 4 Cruise_No = 0284) towed standard
# shrimp trawls (ST) side by side (port/starboard) with experimental trawls (ES). Therefore,
# linking the invrec and bgsrec tabs based on the vessel, cruise_no and p_sta_no variables
# will lead to all catch records for both the shrimp and experimental trawls being linked
# to both trawls. The bgsrec also contains records for catches not associated with invrec table
# records. These are from reef fish cruises. The following codes creates a modified bgsrec table
# that updates the null invrecid for older data and performs some checks.

names(gmex_tow) <- tolower(names(gmex_tow))
names(gmex_bio) <- tolower(names(gmex_bio))
#create bgsrec_invrecid_fix
#get only stationid and invrecid from invrec table
get_stationid_invrecid <- gmex_tow %>% dplyr::select(stationid, invrecid) %>% rename(inv_invrecid = invrecid)

#extract bgsrec table records with missing invrecid and update based on stationid from get_stationid_invrecid
bgsrec_null_invrecid <- gmex_bio %>%
  dplyr::filter(is.na(invrecid)) %>%
  dplyr::left_join(get_stationid_invrecid, by = 'stationid') %>%
  dplyr::mutate(invrecid = inv_invrecid) %>% dplyr::select(-inv_invrecid)

#Extract any remaining bgsrec table records with null invrecid. These should all be
#associated with reef fish cruises at this point.
bgsrec_null_check1 <- bgsrec_null_invrecid %>%
  dplyr::filter(is.na(invrecid))

#extract bgsrec table records with valid invrecid
bgsrec_with_invrecid <- gmex_bio %>%
  dplyr::filter(!is.na(invrecid))

#Stack bgsrec_null_invrecid now updated with valid invrecid and bgsrec_with_invrecid
gmex_bio_mod <- bgsrec_null_invrecid %>%
  dplyr::bind_rows(bgsrec_with_invrecid) %>%
  #Remove null invrecid which should only include those in bgsrec_null_check1
  dplyr::filter(!is.na(invrecid)) %>%
  dplyr::arrange(bgsid)

#Check to make sure only records with invrecs are present - should have 0 rows
bgsrec_null_check2 <- gmex_bio_mod %>% filter(is.na(invrecid))

#Issues 2: Taxonomic coding
# Gmex_bio_mod has a few instances of invalid bio_bgs (biocode) values.
# Also, multiple code/taxonomic combinations may refer to the same organisms under different names.
# Gmex_bio_mod reflects the code/taxonomic use at time of data ingest.
# Gmex_spp will allow translation of cases where multiple code/taxonomic refer to the same organism.
# Since multiple changes may have occurred, the ciu_biocode (currently in use biocode) value ties multiple
# that are now inactive to the current active biocode. Inactive biocodes have the variable inactive set to 1
#
# Notes below from DISMAP (December 2025):

```

```

# (3-1) The newbiocodesbig table does not fully contain all code/taxonomic names found in the bgsrec table
# (3-2) the bgsrec table has a few instances of invalid bio_bgs (biocode) values; and
# (3-3) multiple code/taxonomic combinations may refer to the same organisms under different names. For
# 189040204/MONACANTHUS HISPIDUS, 189040305/STEPHANOLEPIS HISPIDA, 189040306/STEPHANOLEPIS HISPIDUS
# and 189040307/STEPHANOLEPIS HISPIDA (current) have all been used to identify Planehead Filefish due to
# changes in taxonomy. The bgsrec file reflects the code/taxonomic use at time of data ingest.
# The provided master biocode table (MBT) will allow translation of the vast majority cases where multiple
# code/taxonomic refer to the same organism. The process relies on the use of the biocode,
# ciu_biocode and taxon variables in the MBT. The MBT biocode variable in numeric form is equivalent to
# code (character) variable in the newbiocodesbig and bio_bgs (character) variable in the bgsrec tables.
# Similarly the taxon variable in the MBT table is equivalent to taxonomic in the newbiocodesbig table.
# The MBT also has a rb_biocode (replaced by biocode) variable which is the numeric biocode that
# replaces a inactive (inactive = 1 variable) biocode, and allows me to track changes over time.
# Since multiple changes may have occurred, the ciu_biocode (currently in use biocode) value ties multiple
# that are now inactive to the current active biocode. Inactive biocodes have the variable inactive set to 1.
# Using the example above, the ciu_biocode that ties together records of Planehead Filefish is 189040307.
# The following script updates bgsrec table code to ciu_biocode via the MBT table. The rb_biocode variable is
# needed for this purpose.

```

```

# starting with our gmex_bio_mod from above
gmex_bio_utax1 <- gmex_bio_mod %>%
  #convert bgsrec table bio_bgs variable to numeric integer
  dplyr::mutate(bio_bgs = as.integer(bio_bgs)) %>%
  #rename bio_bgs to biocode to allow for easier manipulation with master biocode table (mbt)
  dplyr::rename(biocode = bio_bgs) %>%
  ### take care of Issue 3-2 ###
  # fix invalid zero code and make it the code (999999998) for unidentified specimen
  dplyr::mutate(biocode = ifelse(biocode == 0, 999999998, biocode)) %>%
  # fix invalid unidentified fish code 100000001 to proper code
  dplyr::mutate(biocode = ifelse(biocode == 100000001, 100000000, biocode)) %>%
  # fix invalid unidentified crustacean code 200000001 to proper code
  dplyr::mutate(biocode = ifelse(biocode == 200000001, 200000000, biocode)) %>%
  # fix invalid unidentified crustacean code 300000001 and 300000001 to proper code
  dplyr::mutate(biocode = ifelse(biocode == 300000001, 300000000, biocode)) %>%
  dplyr::mutate(biocode = ifelse(biocode == 300000002, 300000000, biocode)) %>%
  ### take care of Issue 3-3 ###
  #update older inactive biocodes to those currently in use (ciu_biocode)
  dplyr::left_join(dplyr::select(gmex_spp, biocode, taxon, ciu_biocode, taxonomic), by = "biocode") %>%
  #rename taxon to bgs_taxon to keep the original name associated with a biocode
  dplyr::rename(bgs_taxon = taxon) %>%
  #do a left join to bring in taxon associated with ciu_taxon
  dplyr::left_join(dplyr::select(gmex_spp, biocode, taxon), by = c("ciu_biocode" = "biocode"))

### Issue 3: Problematic Taxa with taxonomic issues or problematic separation in the field###
# Collapse taxa with known identification issues and collapse all sponge to single category
# Note this process needs to be implemented after the ciu_biocode update as the statements
# rely on the ciu_biocode. The statements undergo a review with each updated version of the
# MBT (i.e., gmex_spp)
gmex_bio_utax2 <- gmex_bio_utax1 %>%
  #Take care of squid and species complexes...
  #Update the squid genus Loligo and all species under genus Doryteuthis to the genus Doryteuthis
  mutate(ciu_biocode = ifelse(ciu_biocode %in% c(347020200, 347021001, 347021002, 347021003), 347021000, ciu_biocode)) %>%
  mutate(taxon = ifelse(ciu_biocode %in% c(347021000), 'DORYTEUTHIS SP', taxon)) %>%

```

```

# #Update batfish species to Halieutichthys
mutate(ciu_biocode = ifelse(ciu_biocode >= 195050401 & ciu_biocode <= 195050405,195050400,ciu_biocode))
mutate(taxon = ifelse(ciu_biocode %in% c(195050400),'HALIEUTICHTHYS SP',taxon)) %>%
#Update all jellfishy in the genus Aurelia to the genus Aurelia
mutate(ciu_biocode = ifelse(ciu_biocode >= 618010101 & ciu_biocode <= 618010105,618010100,ciu_biocode))
mutate(taxon = ifelse(ciu_biocode %in% c(618010100),'AURELIA',taxon)) %>%
#Update all lionfishes species to the genus Pterois
mutate(ciu_biocode = ifelse(ciu_biocode %in% c(168011901,168011902),168011900,ciu_biocode)) %>%
mutate(taxon = ifelse(ciu_biocode %in% c(168011900),'PTEROIS',taxon)) %>%
#smoothhounds (Mustelus) Managed as species complex, our ids are OK now but in the past assumptions m
mutate(ciu_biocode = ifelse(ciu_biocode %in% c(108031101,108031102,108031103,108031104),108031100,ciu
mutate(taxon = ifelse(ciu_biocode %in% c(108031100),'MUSTELUS SP',taxon)) %>%
#lump all sponge identifications to Porifera
mutate(ciu_biocode = ifelse(ciu_biocode >= 613000000 & ciu_biocode < 616000000,613000000,ciu_biocode))
mutate(taxon = ifelse(ciu_biocode %in% c(613000000),'PORIFERA',taxon)) %>%
#handle out of order Porifera Demospongiae and Agelas and Agelas and Agelasidae in coral numbers
mutate(ciu_biocode = ifelse(ciu_biocode %in% c(999997000,999997020,617170000,617170100),613000000,ciu
mutate(taxon = ifelse(ciu_biocode %in% c(613000000),'PORIFERA',taxon)) %>%
#Collapse all shrimp species in Rimmapenaeus as they are not consistently seperated in the field
mutate(ciu_biocode = ifelse(ciu_biocode %in% c(228012001,228012002),228012000,ciu_biocode)) %>%
mutate(taxon = ifelse(ciu_biocode %in% c(228012000),'RIMAPENAEUS',taxon)) %>%
#Astropecten species have changed, distribution overlap with major east west differences
mutate(biocode = ifelse(biocode >= 691010101 & biocode <= 691010112,691010100,biocode)) %>%
mutate(taxon = ifelse(biocode %in% c(691010100),'ASTROPECTEN',taxon))

## Collapse gmex_bio_ntax2 to have single entry for each taxa for a distinct invrecid (tow)
gmex_bio_ntax3 <- gmex_bio_ntax2 %>%
  group_by(vessel, cruise_no, p_sta_no, cruiseid, stationid, invrecid, ciu_biocode, taxon, bgscod
  summarise(record_cnt = n(),
    # Note: Extrapolated counts (cntexp) & weights (select_bgs) of a taxa for a tow is the sum
    tcntexp = sum(cntexp, na.rm=TRUE),
    tselect_bgs = sum(select_bgs,na.rm=TRUE))

## MERGE the corrected catch/tow/species information from above with cruise information, but only for s
gmex <- left_join(gmex_bio_ntax3, gmex_tow, by = c("cruiseid", "stationid","vessel", "cruise_no", "p_sta
  # add station location and related data
  left_join(gmex_station, by = c("cruiseid", "stationid", "cruise_no", "p_sta_no")) %>%
  # add cruise title
  left_join(gmex_cruise, by = c("cruiseid", "vessel")) %>%
  #filter out YOY (denoted by BSGCODE=T)
  filter(bgscod != "T" | is.na(bgscod))

gmex <- gmex %>% # this takes a couple minutes
  # Trim to high quality SEAMAP trawls, based off the subset used by Jeff Rester's GS_TRAWL_05232011.sa
  # DISMAP trims only to Summer trawls. In contrast, we retain two seasons (summer and fall)
  # this leaves "Summer SEAMAP Groundfish Survey", "Fall SEAMAP Groundfish Survey", "Fall SEAMAP ground
  filter(
    (grepl("Summer", title) | grepl("Fall", title)) &
    grepl('groundfish|Groundfish', title) & # as opposed to plankton or comparative
    # Melissa Karp said in 2025: The previous size of 40 feet wide has been updated to 42 feet wide.
    # However, as of December 2025, the DISMAP code uses 40 ft, so we maintain that here and don't ch

```

```

    gear_size == 40 &
    mesh_size == 1.63 &
    # OP has no letter value
    !grepl("[A-Z]", op)
  ) %>%
  mutate(
    # Create a unique haulid
    haulid = paste(formatC(vessel, width=3, flag=0), formatC(cruise_no, width=3, flag=0), formatC(p_sta, width=3, flag=0)),
    # Extract year where needed
    year = year(mo_day_yr),
    month = month(mo_day_yr),
    day = day(mo_day_yr),
    quarter = case_when(month %in% c(1,2,3) ~ 1,
                        month %in% c(4,5,6) ~ 2,
                        month %in% c(7,8,9) ~ 3,
                        month %in% c(10,11,12) ~ 4),

    season = ifelse(
      grepl("Summer", title), "Summer",
      ifelse(grepl("Fall", title), "Fall", NA
    )),
    # Calculate decimal lat and lon, depth in m, where needed
    s_latd = ifelse(s_latd == 0, NA, s_latd),
    s_lond = ifelse(s_lond == 0, NA, s_lond),
    e_latd = ifelse(e_latd == 0, NA, e_latd),
    e_lond = ifelse(e_lond == 0, NA, e_lond),
    lat = rowMeans(cbind(s_latd + s_latm/60, e_latd + e_latm/60), na.rm=T),
    lon = -rowMeans(cbind(s_lond + s_lonm/60, e_lond + e_lonm/60), na.rm=T),
  )

#add stratum code defined by STAT_ZONE and depth bands (note depth is recorded as m, and depth bands below 21 and 21-60 fathoms))
gmex$depth_zone<-ifelse(gmex$depth_ssta<=36.576, "20",
                        ifelse(gmex$depth_ssta>36.576, "60", NA))
gmex<-gmex %>%
  mutate(stratum = paste(stat_zone, depth_zone, sep= "-"))

# # fix speed
# Trim out or fix speed and duration records
# trim out tows of 0, >60, or unknown minutes
gmex <- gmex %>%
  filter(min_fish <= 60 & min_fish > 0 & !is.na(min_fish )) %>%
  # fix typo according to Jeff Rester: 30 = 3
  mutate(vessel_spd = ifelse(vessel_spd == 30, 3, vessel_spd)) %>%
  # trim out vessel speeds 0, unknown, or >5 (need vessel speed to calculate area trawled)
  filter(vessel_spd <= 5 & vessel_spd > 0 & !is.na(vessel_spd))

# calculate stratum areas
gmex_strats <- gmex %>%
  group_by(stratum) %>%
  summarise(stratumarea = calcarea(lon, lat))
gmex <- left_join(gmex, gmex_strats, by = "stratum")

```



```

# while comsat is still present
# Remove a tow when paired tows exist (same lat/lon/year but different haulid, only Gulf of Mexico)
# identify duplicate tows at same year/lat/lon
dups <- gmex %>%
  group_by(year, lat, lon) %>%
  filter(n() > 1) %>%
  group_by(haulid) %>%
  filter(n() == 1)

# check the comments
dups %>% dplyr::select(p_sta_no, record_cnt, comstat) %>% print(n=100)
# 13 Dec 2025 Malin Pinsky: all 39 are the starboard net of paired trawls. This means we will remove on

# remove the duplicated tows from the dataset
# 13 Dec 2025 Malin Pinsky: the DISMPAP code also removes 3166 rows that mention "PORT" in comstat. Thi
gmex <- gmex %>%
  filter(!haulid %in% dups$haulid
#       & !grepl("PORT", comstat)
  )

gmex <- gmex %>%
  rename(sub_area = source,
        gear = gear_type,
        haul_id = haulid,
        station = stationid,
        verbatim_name = taxon,
        #spp = taxon,
        depth = depth_ssta,
        num = tcntexp,
        wgt = tselect_bgs,
        haul_dur.min = min_fish,
        latitude = lat,
        longitude = lon,
        sbt = temp_bot,
        sst = temp_ssurf
  ) %>%
# adjust for area towed
mutate(
  #haul duration in hours is haul duration minutes * 1 hour/60 minutes
  haul_dur = haul_dur.min/60,
  #area_swept in km^2: knots * 1.8 km/hr/knot * minutes * 1 hr/60 min * width of gear in feet * 0.000
  area_swept = vessel_spd * 1.85200 * haul_dur.min / 60 * gear_size * 0.0003048,
  # kg/km^2
  wgt_cpua = wgt/area_swept,
  # num/km^2
  num_cpua = num/area_swept,
  # kg per 10000m2. calc area trawled in m2: knots * 1.8 km/hr/knot * 1000 m/km * minutes * 1 hr/60 m
  #wtcpue = 10000*wgt/(vessel_spd * 1.85200 * 1000 * haul_dur.min / 60 * gear_size * 0.3048),
  #num_cpue = 1000000*num/(vessel_spd * 1.85200 * 1000 * haul_dur.min / 60 * gear_size * 0.3048),
  # kg per hour
  wgt_cpue = wgt/haul_dur,
  # number per hour

```



```

num_cpue = num/haul_dur,
# add survey column
survey = "GMEX",
country = "United States",
continent = "n_america",
stat_rec = NA,
verbatim_aphia_id = NA,
) %>%
# remove non-fish
filter(
  verbatim_name != '' | !is.na(verbatim_name),
  # remove unidentified spp
  !verbatim_name %in% c('UNID CRUSTA', 'UNID OTHER', 'UNID.FISH', 'CRUSTACEA(INFRAORDER) BRACHYURA',
) %>%
group_by(survey, country, continent, sub_area, stat_rec, verbatim_aphia_id, haul_id, stratum, stratum)
#sum duplicates (all columns the same except for BGSID which
#we don't pull in, and doesn't have any significance other than telling us that
#these are indeed independent observations. we're not sure why this occurs
#in the raw data files, but it was the recommended technique by Jeff in 2012)
summarise(wgt = sumna(wgt),
  num = sumna(num),
  wgt_cpue = sumna(wgt_cpue),
  wgt_cpua = sumna(wgt_cpua),
  num_cpue = sumna(num_cpue),
  num_cpua = sumna(num_cpua)
) %>%
#add region column
mutate(region = "Gulf of Mexico") %>%
# note this doesn't retain sst or sbt columns
dplyr::select(survey, haul_id, country, sub_area, continent, stat_rec, station, stratum,
  year, month, day, quarter, season, latitude, longitude, haul_dur, area_swept,
  gear, depth, sst, sbt, verbatim_name, num, num_cpua, num_cpue,
  wgt, wgt_cpua, wgt_cpue, verbatim_aphia_id) %>%
ungroup()

#-----#
#### INTEGRATE CLEAN TAXA FROM TAXA ANALYSIS ####
#-----#

# Get WoRM's id for sourcing
worm <- gna_data_sources() %>%
  filter(title == "World Register of Marine Species") %>%
  pull(id)

### Automatic cleaning
# Set Survey code
gmex_survey_code <- "GMEX"

gmex <- gmex %>%
  mutate(
    taxa2 = str_squish(verbatim_name),

```

```

    taxa2 = str_remove_all(taxa2," spp.| sp.| spp| sp|NO "),
    taxa2 = str_to_sentence(str_to_lower(taxa2))
  )

# Get clean taxa
clean_auto <- clean_taxa(unique(gmex$taxa2),
                        input_survey = gmex_survey_code,
                        save = F, output=NA, fishbase=T)
# [1] "Returned 911 taxa and dropped 1431. Misspelled taxa: 109; No alphia id found: 136; Non-fish clas
# Time difference of -2.786229 mins (13 December 2025, Malin Pinsky)

#previous queries missed Astroscopus y-graecum, but not now (13 Dec 2025). no longer need to add it man
clean_auto.missing <- clean_auto

#-----#
#### INTEGRATE CLEAN TAXA in GMEX survey data ####
#-----#
clean_taxa <- clean_auto.missing %>%
  dplyr::select(-survey)

clean_gmex <- left_join(gmex, clean_taxa, by=c("taxa2"="query")) %>%
  filter(!is.na(taxa)) %>%
  # query does not indicate taxa entry that were removed in the cleaning procedure
  # so all NA taxa have to be removed from the surveys because:
  #non-existing, non marine or non fish
  rename(accepted_name = taxa,
         aphia_id = worms_id) %>%
  mutate(verbatim_aphia_id = NA,
         source = "NOAA",
         timestamp = lubridate::my("12/2025"), # date for this dataset creation
         survey_unit = ifelse(survey %in% c("BITS", "NS-IBTS", "SWC-IBTS"),
                              paste0(survey, "-", quarter), survey),
         survey_unit = ifelse(survey %in% c("NEUS", "SEUS", "SCS", "GMEX"),
                              paste0(survey, "-", season), survey_unit)) %>%
  # trim to the FISHGLOB standard columns
  dplyr::select(fishglob_data_columns$`Column name fishglob`)

#check for duplicated accepted_names in the same haul_id
count_clean_gmex <- clean_gmex %>%
  group_by(haul_id, accepted_name) %>%
  mutate(count = n())

#which ones are duplicated? they will be multiple lines that have the same accepted_name
unique_name_match <- count_clean_gmex %>%
  group_by(accepted_name, verbatim_name) %>%
  filter(count>1) %>%
  distinct(accepted_name, verbatim_name)
unique_name_match
# 13 Dec 2025, Malin Pinsky: explanations for duplications of haulid x species
# EUCINOSTOMUS ARGENTEUS/GULA, EUCINOSTOMUS HARENGULUS/JONESII, and EUCINOSTOMUS all fix to Eucinostomus
# CYNOSCIION and CYNOSCIION ARENARIUS/NOTHUS both resolve to Cynoscion (a genus record)

```

```

# We will leave it to users to decide what to do with these repeats due to taxonomic classifications
# It will depend on goals of data use. We therefore maintain these as multiple entries in the FishGlob

# check for duplicate with the same species and the same data (wgt,num)
clean_gmex %>% group_by(haul_id, accepted_name, wgt, num) %>% filter(n()>1)
# 13 Dec 2025 Malin Pinsky: none. good!

##### A. Fredston, August 2025: resolving issue #49 where haul_id value is a numeric, see https://
class(clean_gmex$haul_id)
head(clean_gmex$haul_id)

clean_gmex_fixed_haul_id <- clean_gmex |>
  mutate(haul_id = paste0("id", haul_id))
class(clean_gmex_fixed_haul_id$haul_id)
head(clean_gmex_fixed_haul_id$haul_id)
#####

# -----#
#### SAVE DATABASE ####
# -----#

# Just run this routine should be good for all
write_clean_data(data = clean_gmex_fixed_haul_id, survey = "GMEX", overwrite = T, csv = T)

# -----#
#### FLAGS ####
# -----#

#install required packages that are not already installed
required_packages <- c("data.table",
  "devtools",
  "dggridR",
  "dplyr",
  "fields",
  "forcats",
  "ggplot2",
  "here",
  "magrittr",
  "maps",
  "maptools",
  "raster",
  "rcompendium",
  "readr",
  "remotes",
  "rrtools",
  "sf",
  "sp",
  "tidyr",
  "usethis")

not_installed <- required_packages[!(required_packages %in% installed.packages()[ , "Package"])]

```

```

if(length(not_installed)) install.packages(not_installed)

#load pipe operator
library(magrittr)

##### Apply taxonomic flagging per region
#get vector of regions (here the survey column)
regions <- levels(as.factor(clean_gmex$survey))

#run flag_spp function in a loop
for (r in regions) {
  flag_spp(clean_gmex, r)
}

##### Apply trimming per survey_unit method 1
#apply trimming for hex size 7
dat_new_method1_hex7 <- apply_trimming_per_survey_unit_method1(clean_gmex, 7)

#apply trimming for hex size 8
dat_new_method1_hex8 <- apply_trimming_per_survey_unit_method1(clean_gmex, 8)

##### Apply trimming per survey_unit method 2
dat_new_method2 <- apply_trimming_per_survey_unit_method2(clean_gmex)

#-----#
#### ADD STRANDARDIZATION FLAGS ####
#-----#
surveys <- sort(unique(clean_gmex$survey))
survey_units <- sort(unique(clean_gmex$survey_unit))
survey_std <- clean_gmex %>%
  mutate(flag_taxa = NA_character_,
         flag_trimming_hex7_0 = NA_character_,
         flag_trimming_hex7_2 = NA_character_,
         flag_trimming_hex8_0 = NA_character_,
         flag_trimming_hex8_2 = NA_character_,
         flag_trimming_2 = NA_character_)

# integrate taxonomic flags
for(i in 1:length(surveys)){
  if(!surveys[i] %in% c("FALK","GSL-N","MRT","NZ-CHAT","SCS", "SWC-IBTS")){
    xx <- data.frame(read_delim(paste0("outputs/Flags/taxonomic_flagging/",
                                       surveys[i], "_flagspp.txt"),
                              delim=";", escape_double = FALSE, col_names = FALSE,
                              trim_ws = TRUE))
    xx <- as.vector(unlist(xx[1,]))

    survey_std <- survey_std %>%
      mutate(flag_taxa = ifelse(survey == surveys[i] & accepted_name %in% xx,
                              "TRUE",flag_taxa))

    rm(xx)
  }
}

```

```

}
}

# integrate spatio-temporal flags
for(i in 1:length(survey_units)){

  if(!survey_units[i] %in% c("DFO-SOG", "IS-TAU", "SCS-FALL", "WBLS")){

    hex_res7_0 <- read.csv(paste0("outputs/Flags/trimming_method1/hex_res7/",
                                  survey_units[i], "_hex_res_7_trimming_0_hauls_removed.csv"),
                          sep = ";", colClasses=c(haul_id = "character"))
    hex_res7_0 <- as.vector(hex_res7_0[,1])

    hex_res7_2 <- read.csv(paste0("outputs/Flags/trimming_method1/hex_res7/",
                                  survey_units[i], "_hex_res_7_trimming_02_hauls_removed.csv"),
                          sep = ";", colClasses=c(haul_id = "character"))
    hex_res7_2 <- as.vector(hex_res7_2[,1])

    hex_res8_0 <- read.csv(paste0("outputs/Flags/trimming_method1/hex_res8/",
                                  survey_units[i], "_hex_res_8_trimming_0_hauls_removed.csv"),
                          sep= ";", colClasses=c(haul_id = "character"))
    hex_res8_0 <- as.vector(hex_res8_0[,1])

    hex_res8_2 <- read.csv(paste0("outputs/Flags/trimming_method1/hex_res8/",
                                  survey_units[i], "_hex_res_8_trimming_02_hauls_removed.csv"),
                          sep = ";", colClasses=c(haul_id = "character"))
    hex_res8_2 <- as.vector(hex_res8_2[,1])

    trim_2 <- read.csv(paste0("outputs/Flags/trimming_method2/",
                              survey_units[i], "_hauls_removed.csv"), colClasses=c(haul_id_removed = "character"))
    trim_2 <- as.vector(trim_2[,1])

    survey_std <- survey_std %>%
      mutate(flag_trimming_hex7_0 = ifelse(survey_unit == survey_units[i] & haul_id %in% hex_res7_0,
                                           "TRUE", flag_trimming_hex7_0),
             flag_trimming_hex7_2 = ifelse(survey_unit == survey_units[i] & haul_id %in% hex_res7_2,
                                           "TRUE", flag_trimming_hex7_2),
             flag_trimming_hex8_0 = ifelse(survey_unit == survey_units[i] & haul_id %in% hex_res8_0,
                                           "TRUE", flag_trimming_hex8_0),
             flag_trimming_hex8_2 = ifelse(survey_unit == survey_units[i] & haul_id %in% hex_res8_2,
                                           "TRUE", flag_trimming_hex8_2),
             flag_trimming_2 = ifelse(survey_unit == survey_units[i] & haul_id %in% trim_2,
                                      "TRUE", flag_trimming_2)
      )
    rm(hex_res7_0, hex_res7_2, hex_res8_0, hex_res8_2, trim_2)
  }
}

# verify that the flagging worked. these values should match the respective _stats_hauls.csv files in o
survey_std |>
group_by(survey_unit) |>
distinct(haul_id, flag_trimming_hex7_0, flag_trimming_hex7_2, flag_trimming_hex8_0, flag_trimming_hex8_2, flag_trimming_2)
summarize(hex7_0 = sum(!is.na(flag_trimming_hex7_0)),

```

```

hex7_2 = sum(!is.na(flag_trimming_hex7_2)),
hex8_0 = sum(!is.na(flag_trimming_hex8_0)),
hex8_2 = sum(!is.na(flag_trimming_hex8_2)),
trim_2 = sum(!is.na(flag_trimming_2))

##### A. Fredston, August 2025: resolving issue #49 where haul_id value is a numeric, see https://
class(survey_std$haul_id)
head(survey_std$haul_id)

survey_std_fixed_haul_id <- survey_std |>
  mutate(haul_id = paste0("id", haul_id))
class(survey_std_fixed_haul_id$haul_id)
head(survey_std_fixed_haul_id$haul_id)
#####

# Just run this routine should be good for all
write_clean_data(data = survey_std_fixed_haul_id, survey = "GMEX_std",
  overwrite = T, rdata=TRUE)

```


1. Overview of the survey data table

survey	source	timestamp	haul_id	country	sub_area	continent
GMEX	NOAA	2025-12-01	id023-011- 2301	United States	AL	n_america
GMEX	NOAA	2025-12-01	id023-011- 2302	United States	AL	n_america
GMEX	NOAA	2025-12-01	id023-011- 2302	United States	AL	n_america
GMEX	NOAA	2025-12-01	id023-011- 2302	United States	AL	n_america
GMEX	NOAA	2025-12-01	id023-011- 2302	United States	AL	n_america

stat_rec	station	stratum	year	month	day	quarter	season
NA	58381	11-20	2001	6	18	2	Summer
NA	58382	11-20	2001	6	18	2	Summer
NA	58382	11-20	2001	6	18	2	Summer
NA	58382	11-20	2001	6	18	2	Summer
NA	58382	11-20	2001	6	18	2	Summer

latitude	longitude	haul_dur	area_swept	gear	depth
30.15125	-88.052	0.1666667	0.0112898	ST	9.1
30.03933	-88.404	0.2666667	0.0180637	ST	20.1
30.03933	-88.404	0.2666667	0.0180637	ST	20.1
30.03933	-88.404	0.2666667	0.0180637	ST	20.1
30.03933	-88.404	0.2666667	0.0180637	ST	20.1

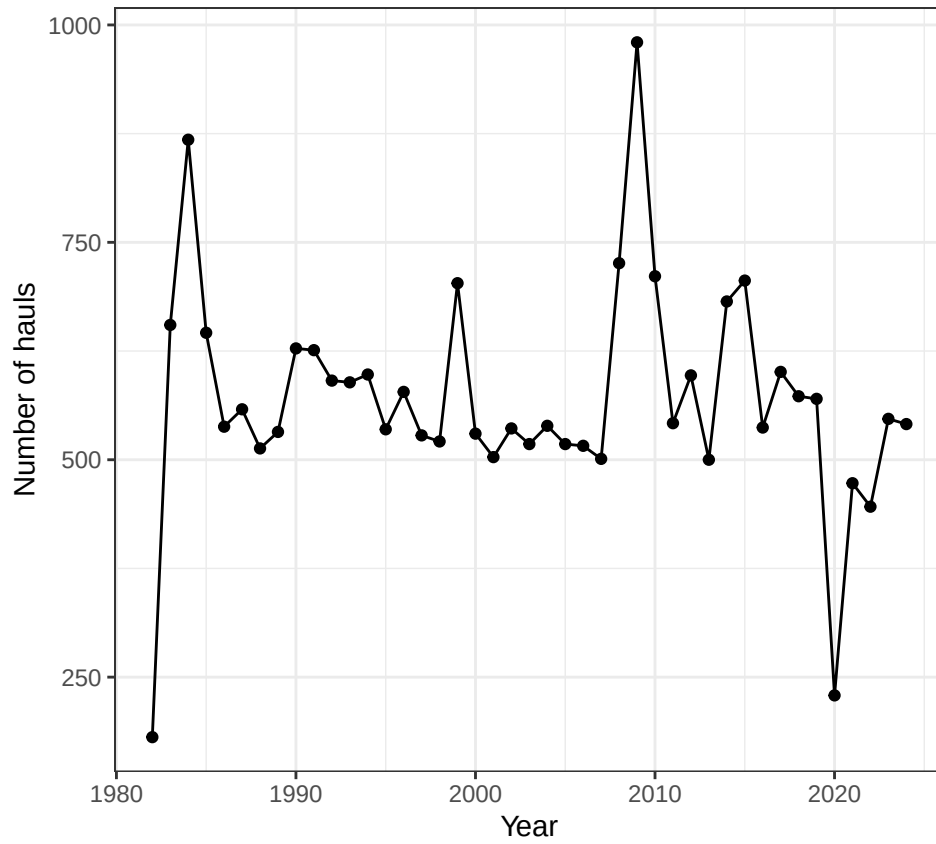
sbt	sst	num	num_cpue	num_cpua	wgt
22.63	27.83	1	6.00	88.57559	0.012
21.85	28.35	5	18.75	276.79872	0.032
21.85	28.35	1	3.75	55.35974	0.015
21.85	28.35	1	3.75	55.35974	0.016
21.85	28.35	3	11.25	166.07923	0.022

wgt_cpue	wgt_cpua	verbatim_name	verbatim_aphia_id	accepted_name
0.07200	1.0629071	CENTROPRISTIS PHILADELPHICA	NA	Centropristis philadelphica
0.12000	1.7715118	ETROPUS CROSSOTUS	NA	Etropus crossotus
0.05625	0.8303962	LAGOCEPHALUS LAEVIGATUS	NA	Lagocephalus laevigatus
0.06000	0.8857559	PRIONOTUS RUBIO	NA	Prionotus rubio
0.08250	1.2179144	SAURIDA BRASILIENSIS	NA	Saurida brasiliensis

aphia_id	SpecCode	kingdom	phylum	class	order	family
159347	3317	Animalia	Chordata	Teleostei	Perciformes	Serranidae
158799	4221	Animalia	Chordata	Teleostei	Pleuronectiformes	Paralichthyidae
158933	1239	Animalia	Chordata	Teleostei	Tetraodontiformes	Tetraodontidae
276289	4025	Animalia	Chordata	Teleostei	Perciformes	Triglidae
158756	2716	Animalia	Chordata	Teleostei	Aulopiformes	Synodontidae

2. Summary of sampling intensity

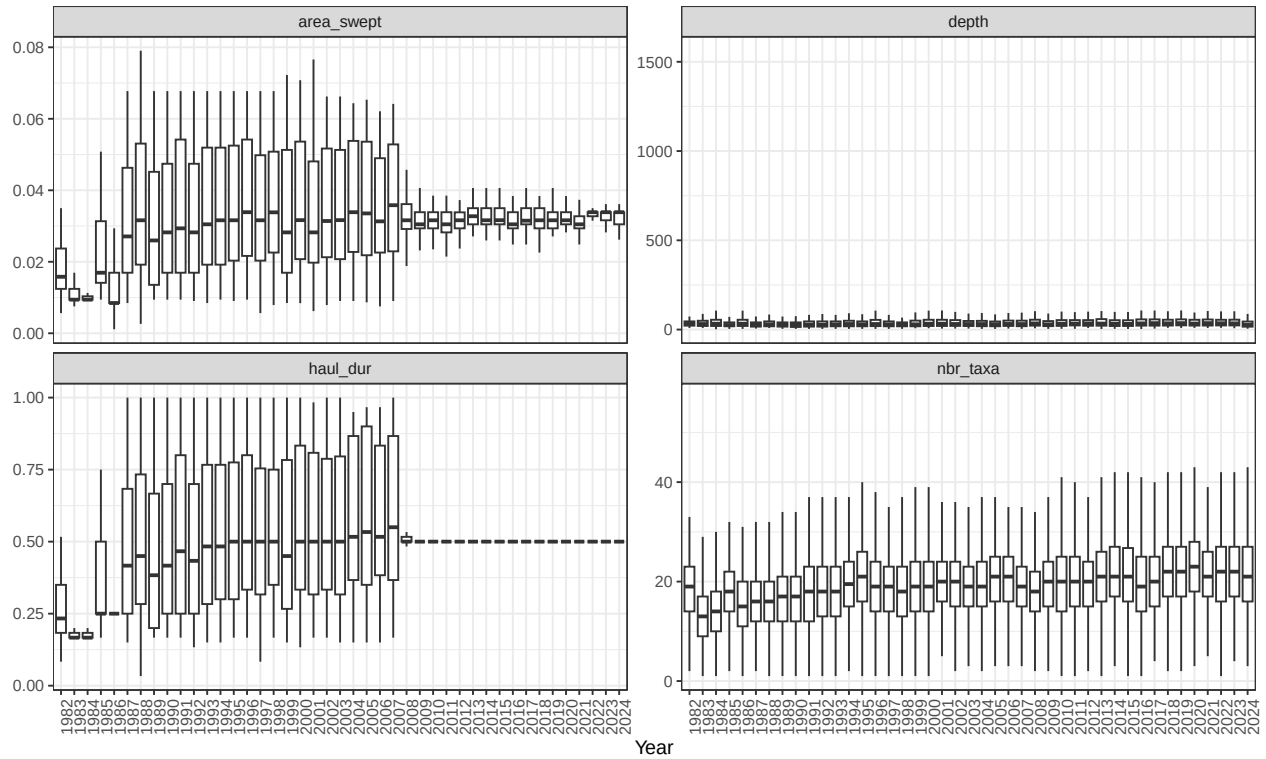
Number of hauls per year performed during the survey after data processing.



3. Summary of sampling variables from the survey

Here we show the yearly total and average of the following variables reported in the survey data:

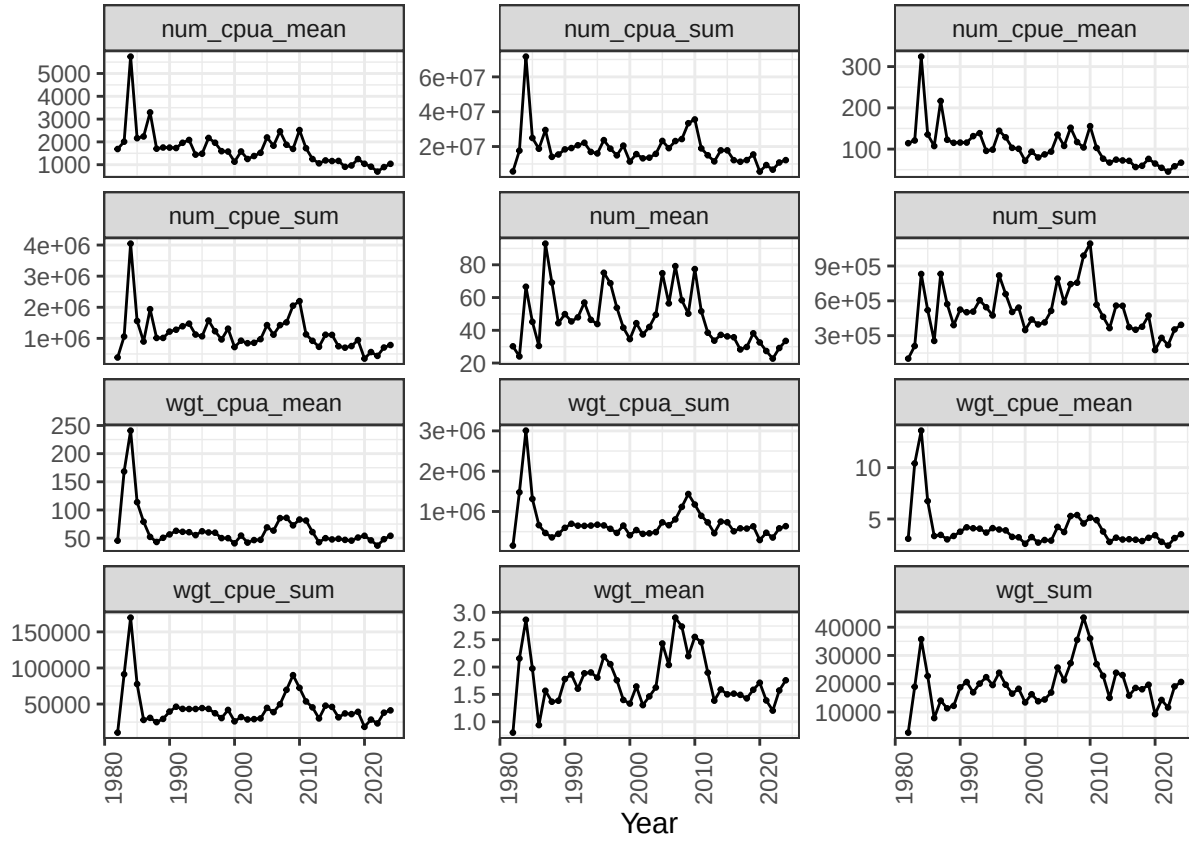
- *area_swept*, swept area by the bottom trawl gear km^2
- *depth*, sampling depth in *m*
- *haul_dur*, haul sampling duration *hour*
- *number of marine fish taxa*, taxa were cleaned following the last version of taxonomy from the World Register of Marine Species (<https://www.marinespecies.org/>, October 2021)



4. Summary of biological variables

Here we display the yearly total and average across hauls of the following variables recorded in the data:

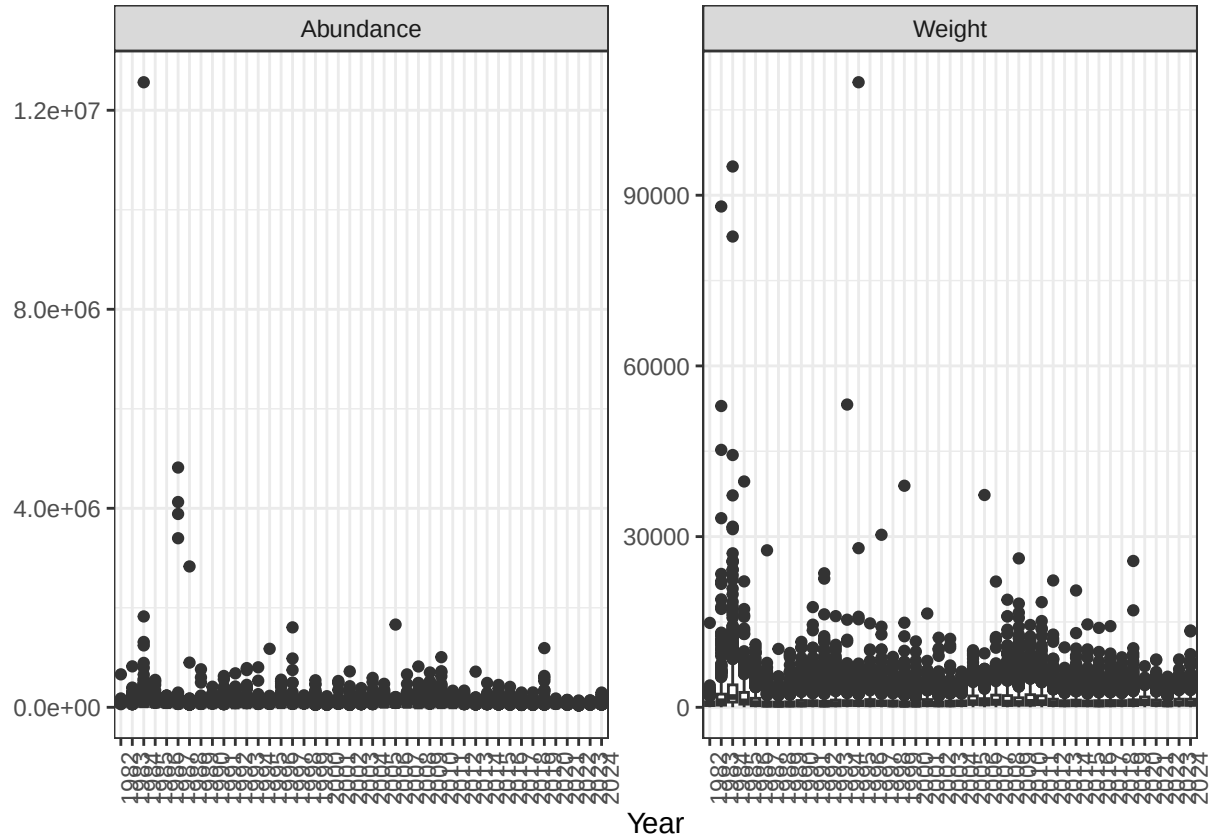
- *num_cpua*, number of individuals (abundance) in $\frac{\text{individuals}}{\text{km}^2}$
- *num_cpue*, number of individuals (abundance) in $\frac{\text{individuals}}{h}$
- *num*, number of individuals (abundance)
- *wgt_cpua*, weight in $\frac{kg}{\text{km}^2}$
- *wgt_cpue*, weight in $\frac{kg}{h}$
- *wgt*, weight in *kg*



5. Extreme values

Here we show a yearly total distribution of the biomass data to visualize outliers:

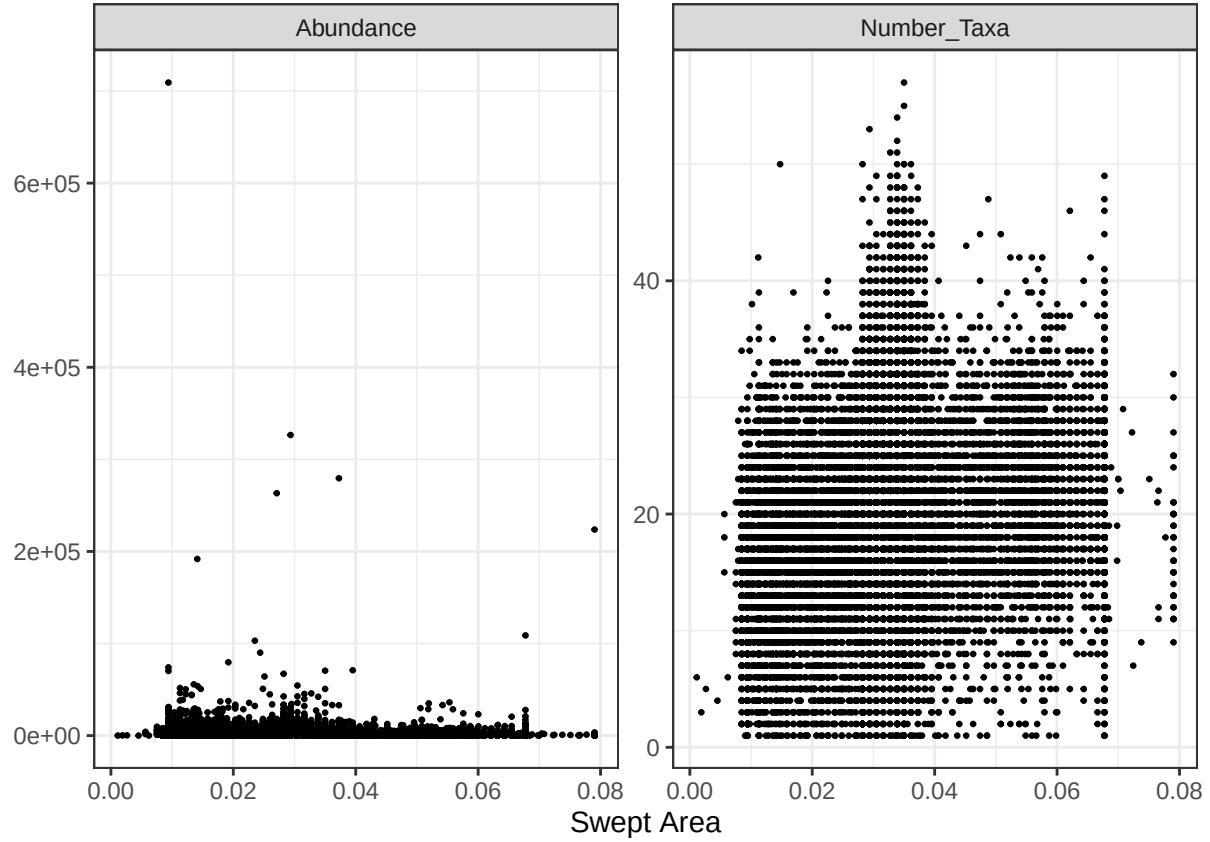
- *num_cpue*, number of individuals (abundance) in $\frac{\text{individuals}}{\text{km}^2}$
- *wgt_cpue*, weight in $\frac{\text{kg}}{\text{km}^2}$



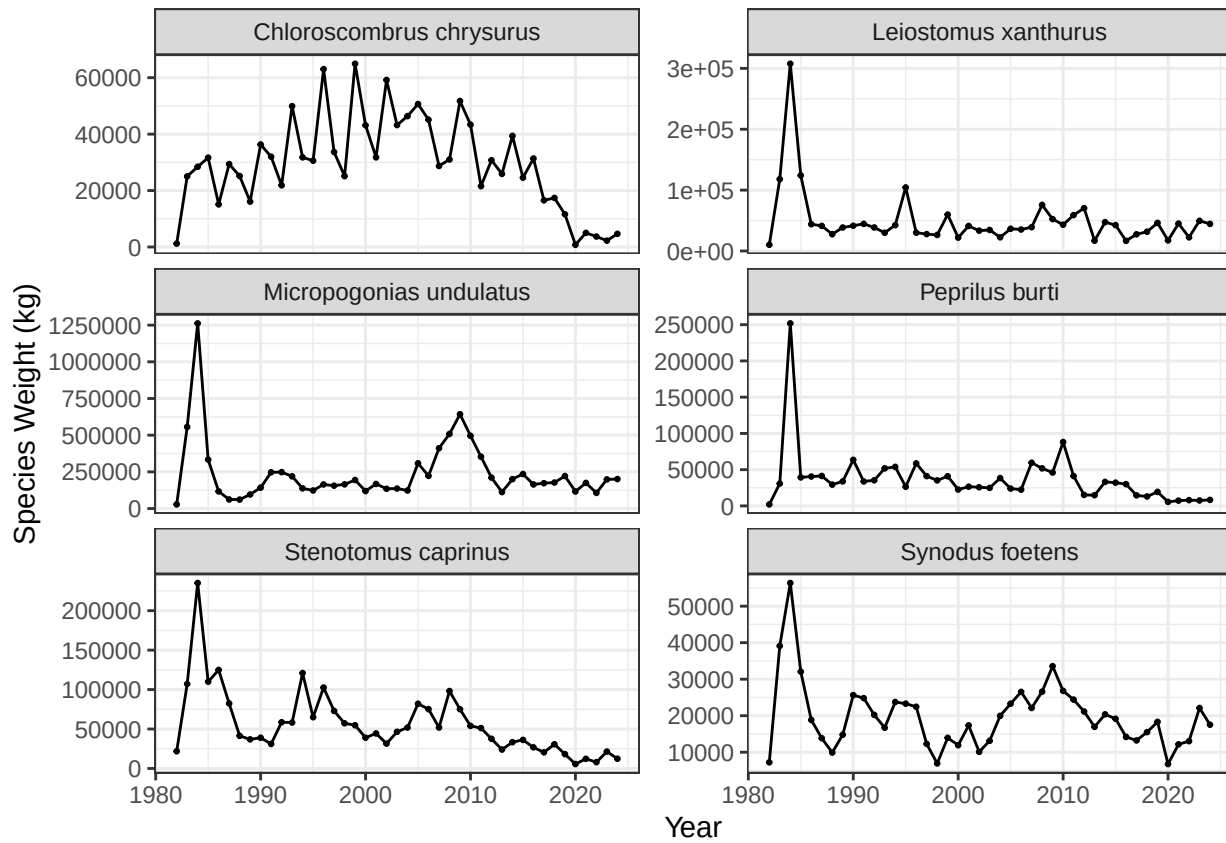
6. Summary of variables against swept area

Here we show the total abundance and number of taxa relationships with the area swept:

- *nbr_taxa*, number of marine fish taxa after taxonomic data cleaning
- *num_cpua*, number of individuals (abundance) in $\frac{\text{individuals}}{\text{km}^2}$
- *wgt_cpua*, weight in $\frac{\text{kg}}{\text{km}^2}$

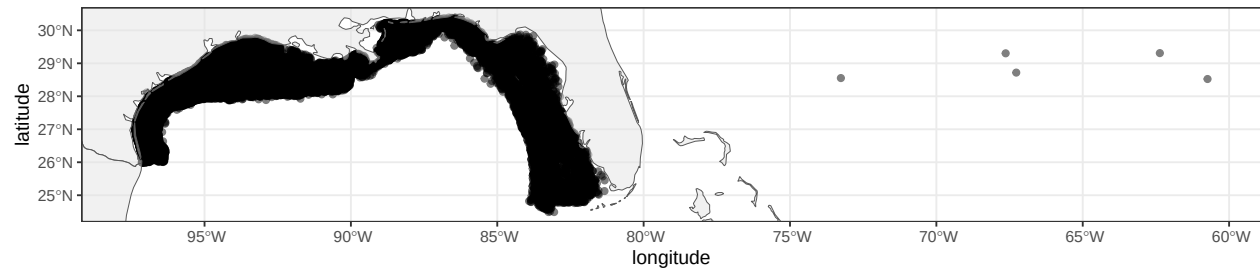


7. Abundance or Weight trends of the six most abundant species



8. Distribution mapping

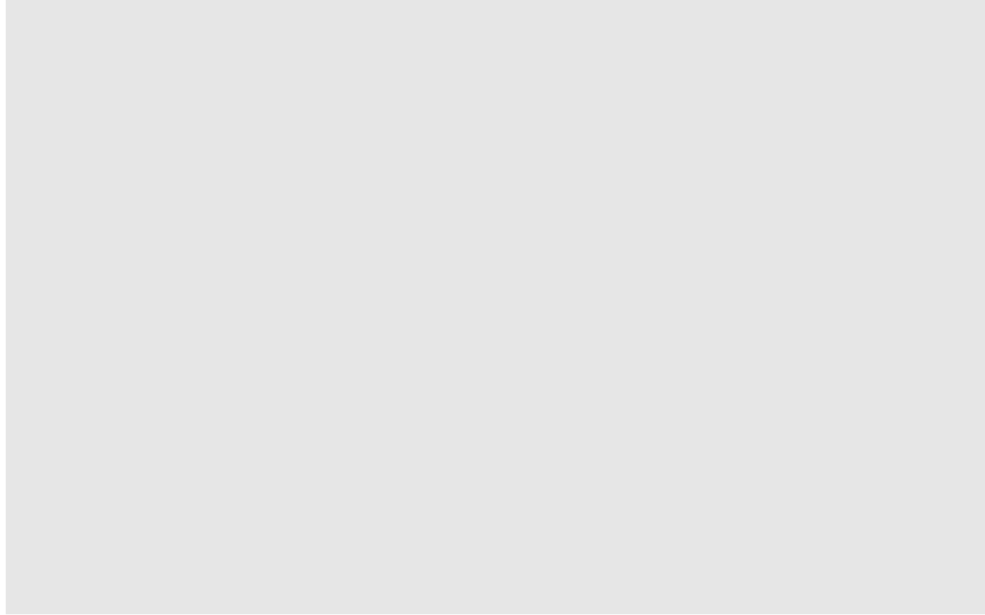
Map of the sampling distribution in space. Note that we only show one year per coordinate.



9. Taxonomic flagging

This species flagging method was adapted from <https://github.com/pinskylab/OceanAdapt/blob/master/R/add-spp-to-taxonomy.Rmd#L33>

Visualization of flagged taxa



Statistics related to the taxonomic flagging outputs

Total number of species	906
Percentage of species flagged	0

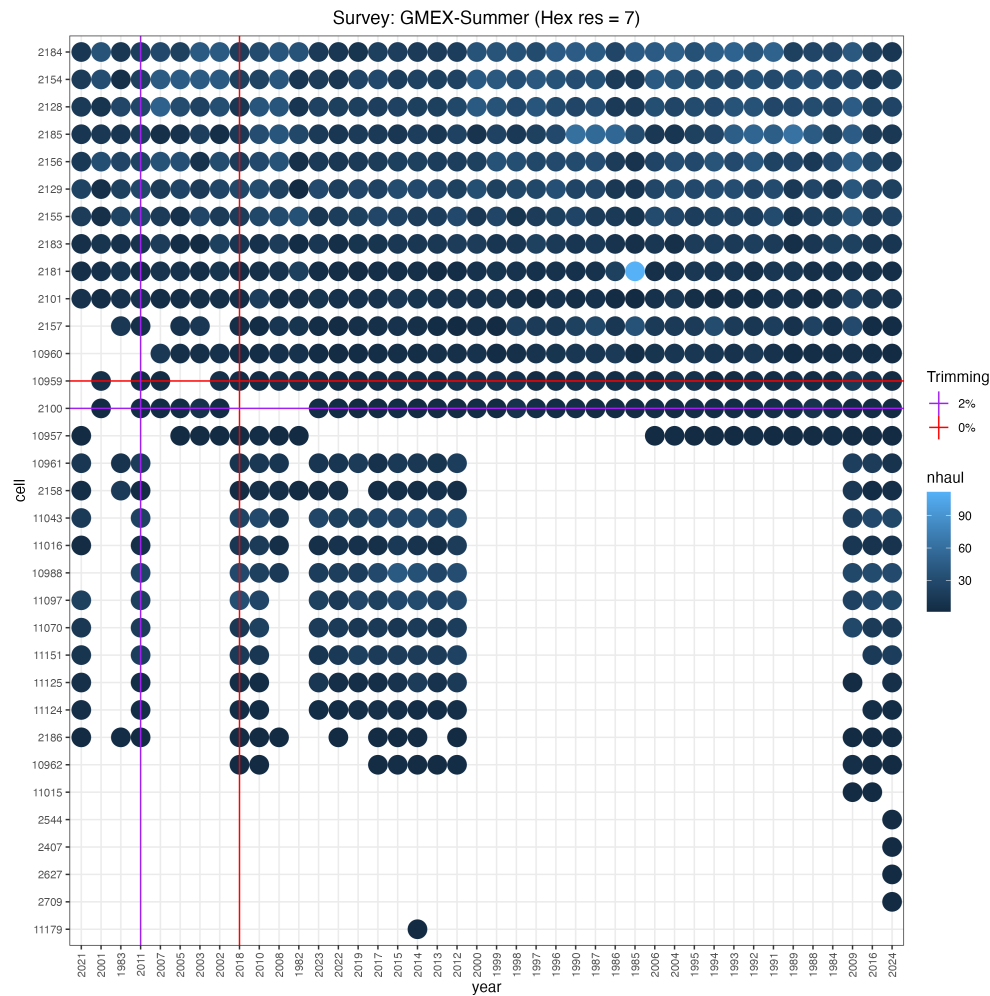
10. Spatio-temporal standardization: GMEX-Summer

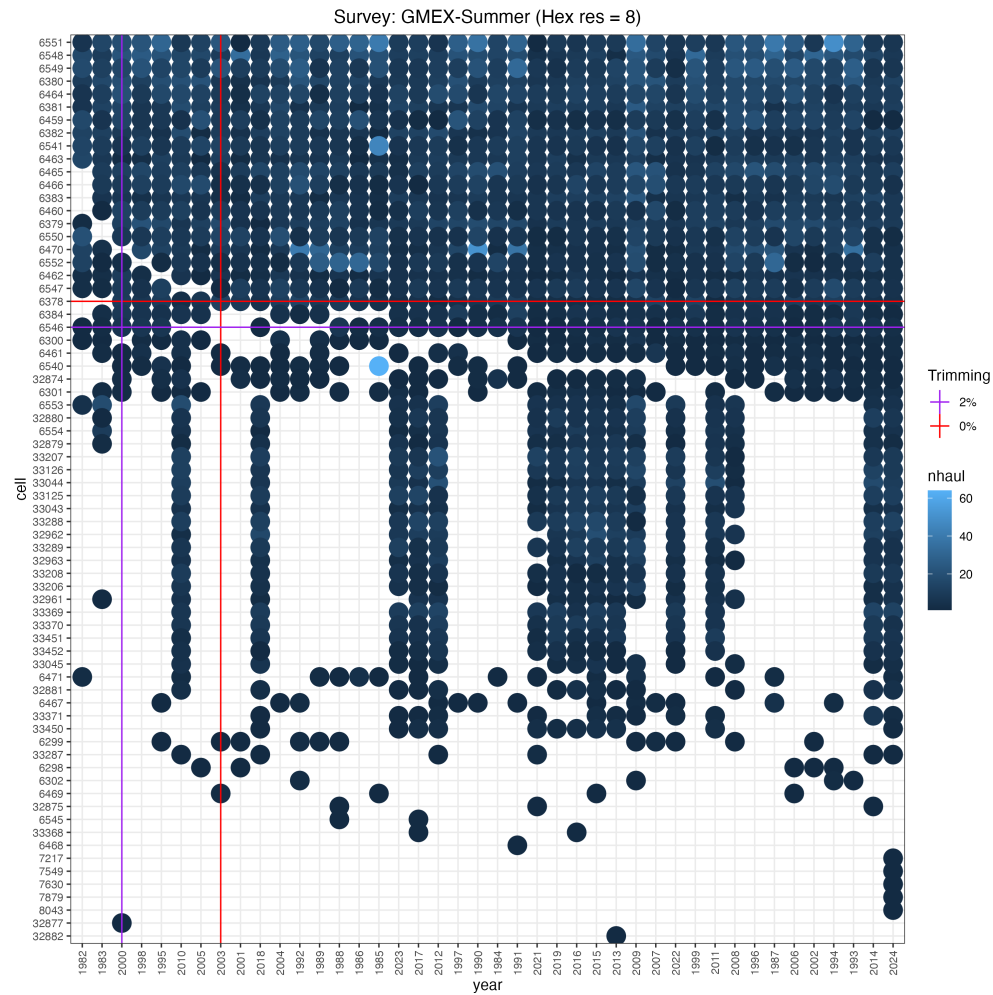
a. Standardization method 1

This standardization method was adapted from https://github.com/zoekitchel/rawl_spatial_turnover/blob/master/data_prep_code/species/explore_NorthSea_trimming.Rmd

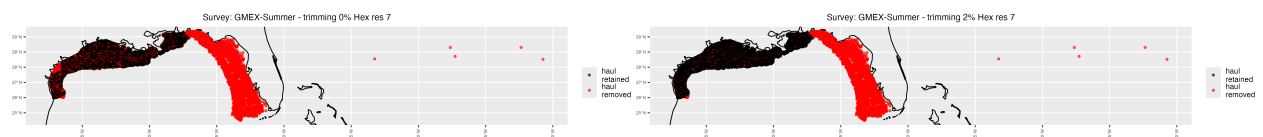
It was run for hex resolution 7 and 8.

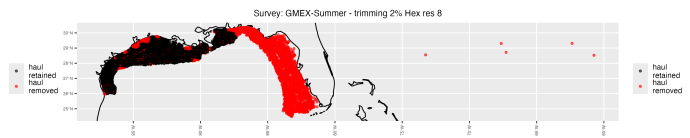
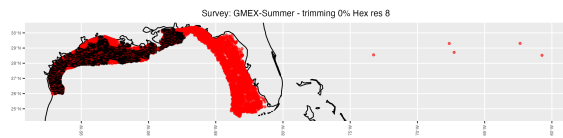
Plot of number of cells x years with overlaid flagging options



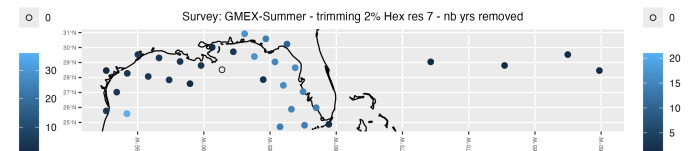
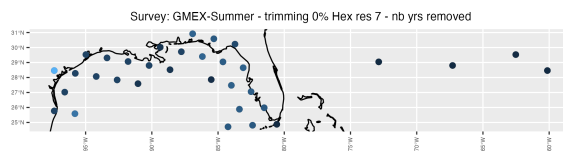


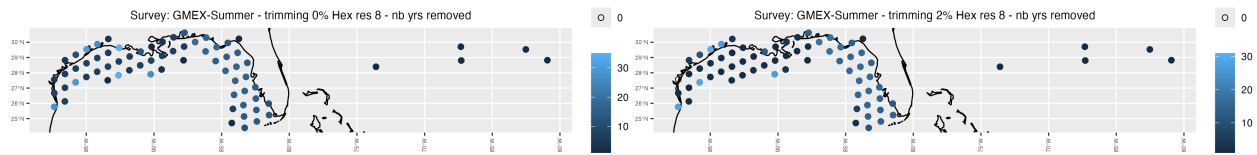
Map of hauls retained and removed per flagging method and threshold





Map of numbers of years removed per grid cell and flagging method/threshold

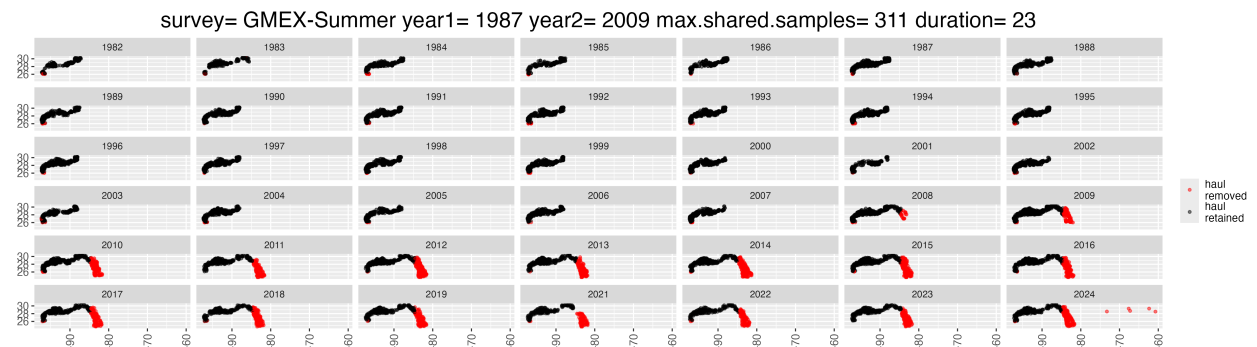




b. Standardization method 2

This standardization method was adapted from BioTIME code from https://github.com/Wubing-Xu/Range_size_winners_losers

Map of hauls retained and removed



c. Standardization summary

Statistics of hauls removed for each standardization method

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.

## Warning in styling_latex_position_float(x, table_info, "l", table.envir, :
## wraptable is not supported for longtable.
```

summary	grid cell 7, 0% threshold	grid cell 7, 2% threshold	grid cell 8, 0% threshold	grid cell 8, 2% threshold
number of hauls removed	3933.0	2746.0	4455	3087
percentage of hauls removed	32.7	22.8	37	25.5

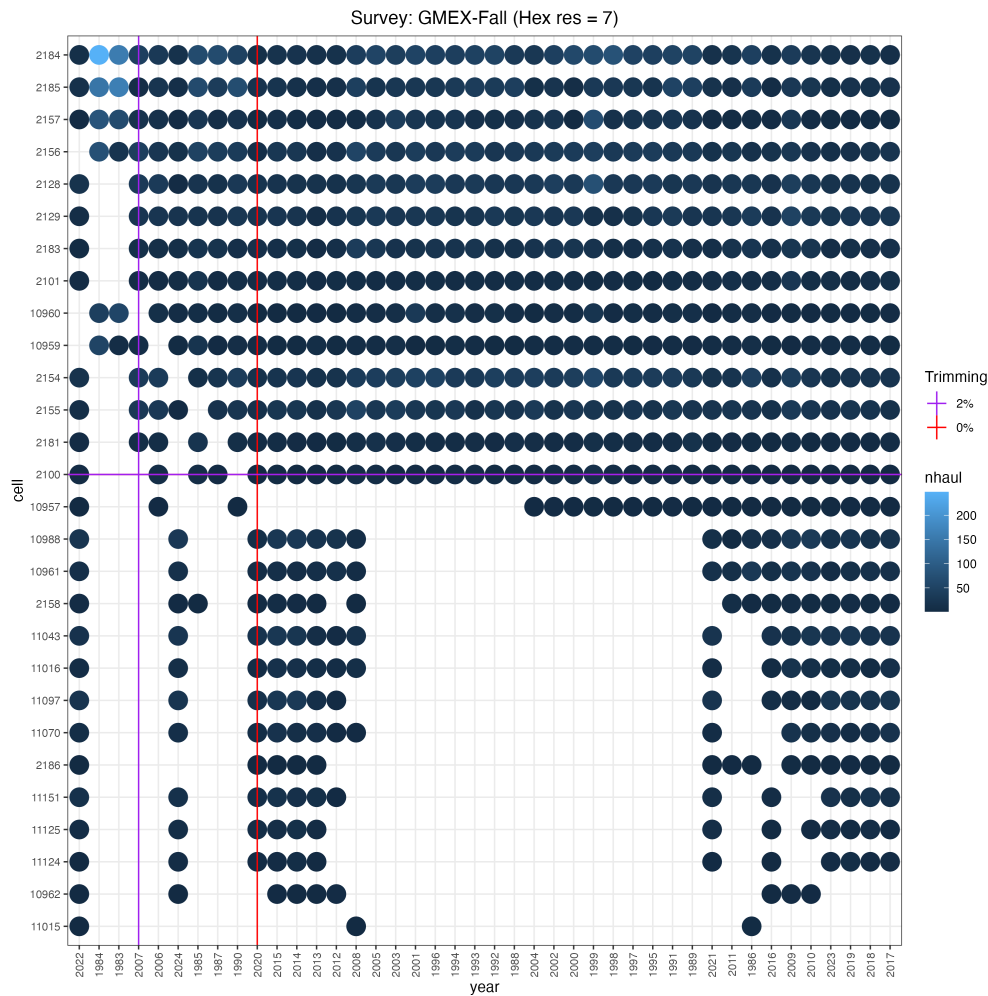
11. Spatio-temporal standardization: GMEX-Fall

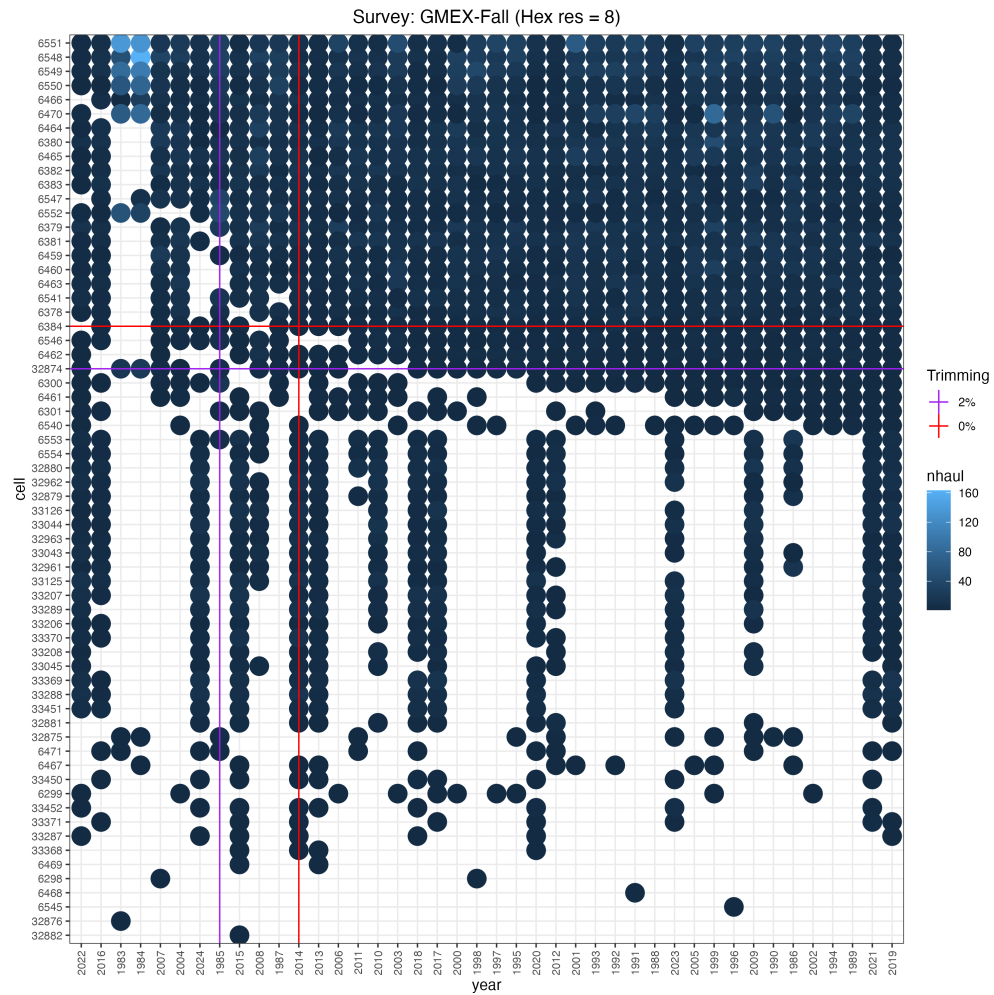
a. Standardization method 1

This standardization method was adapted from https://github.com/zoekitchel/rawl_spatial_turnover/blob/master/data_prep_code/species/explore_NorthSea_trimming.Rmd

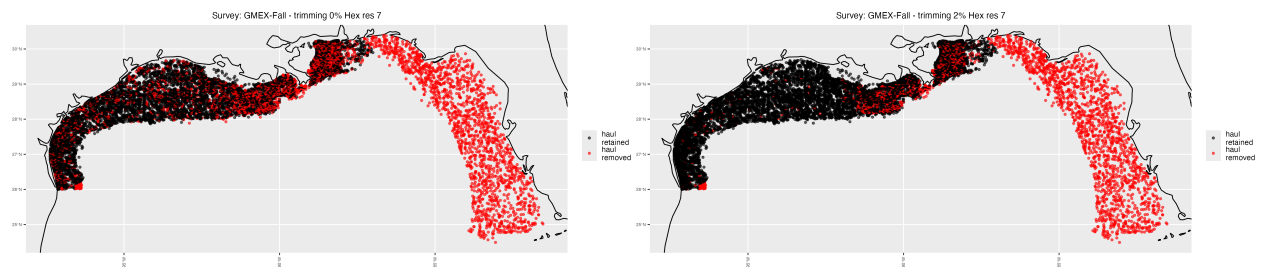
It was run for hex resolution 7 and 8.

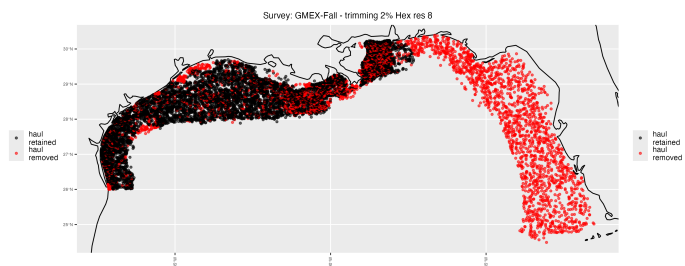
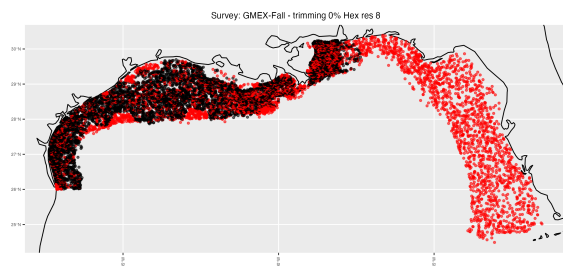
Plot of number of cells x years with overlaid flagging options



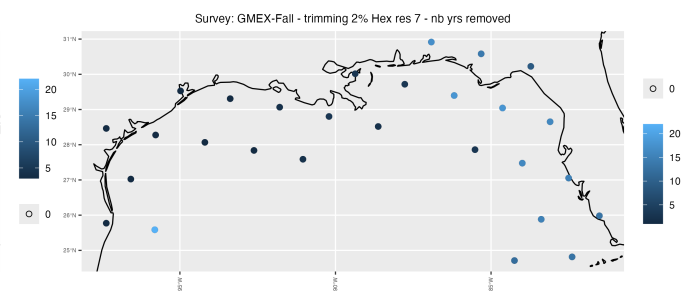
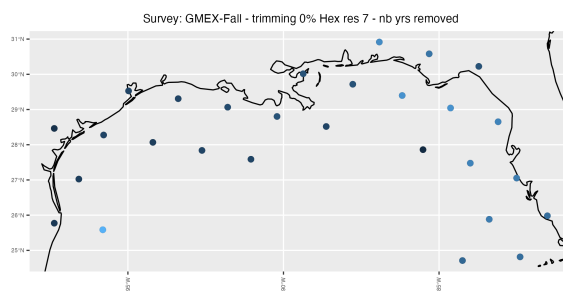


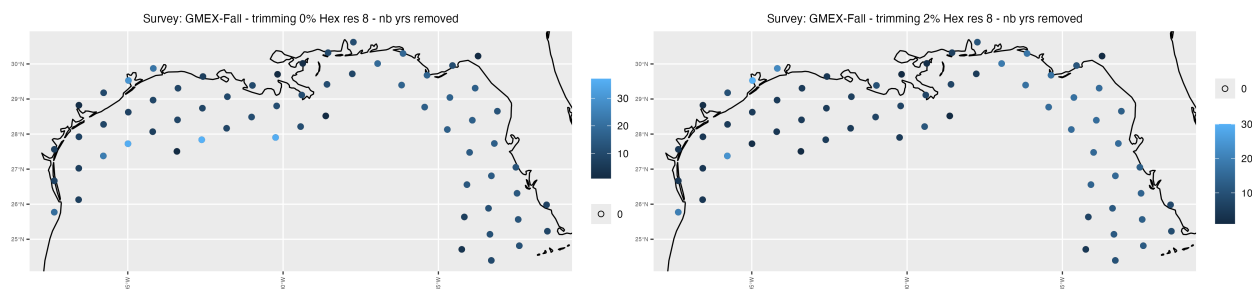
Map of hauls retained and removed per flagging method and threshold





Map of numbers of years removed per grid cell and flagging method/threshold

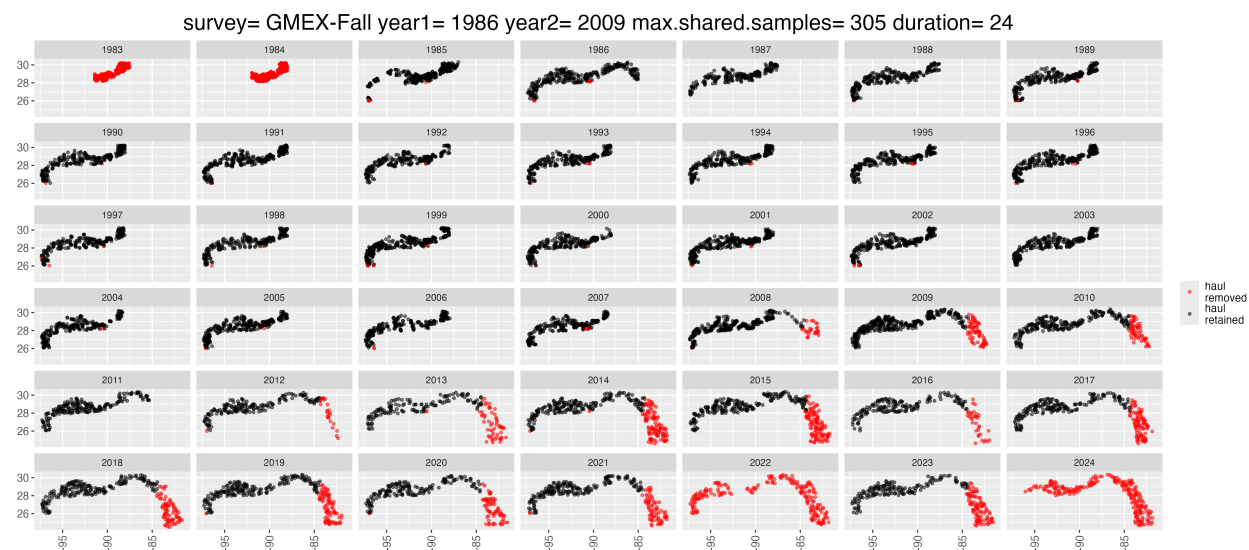




b. Standardization method 2

This standardization method was adapted from BioTIME code from https://github.com/Wubing-Xu/Range_size_winners_losers

Map of hauls retained and removed



c. Standardization summary

Statistics of hauls removed for each standardization method

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

```
## Warning in styling_latex_position_float(x, table_info, "l", table.envir, :
## wraptable is not supported for longtable.
```

summary	grid cell 7, 0% threshold	grid cell 7, 2% threshold	grid cell 8, 0% threshold	grid cell 8, 2% threshold
number of hauls removed	4439.0	2937.0	5254.0	3454.0
percentage of hauls removed	35.6	23.5	42.1	27.6